Lecture 3 Exercise Solutions

1. Find the multiplicative inverse of $7 \bmod 59$ by computing $7^{57} \bmod 59$. Explain why it works.

   59 is prime. Therefore by Fermat's Little Theorem $7^{58} \equiv 1 \bmod 59$, or $\bar{7}^{58} = \bar{1}$. But, $\bar{7}^{58} = \bar{7} \cdot 7^{57}$, so $\bar{7} \cdot 7^{57} = \bar{1} \Rightarrow \bar{7}^{57}$ is the inverse of $\bar{7}$, so we have to find $7^{57} \bmod 59$.

X	e	Y
7	57	1
7	56	7
49	28	7
41	14	7
29	7	7
29	6	26
15	3	26
15	2	36
48	1	36
48	0	17

   So, $7^{57} \bmod 59 = 17$ and $\bar{7}^{-1} = \overline{17} \bmod 59$. Indeed, $7 \cdot 17 = 119$ and $119 = 2 \cdot 59 + 1$.

   <u>Generalization of this example</u>:
   Find the multiplicative inverse of $\bar{a}$ in $Z_m^{\times}$ or in other words, find the multiplicative inverse of a mod m, where m is a prime number. By Fermat's Little Theorem $a^{m-1} \equiv 1 \bmod m$. But, $a^{m-1} \bmod m = a \cdot a^{m-2} \bmod m = 1$. Thus, $a^{m-2} \bmod m$ is the inverse of a. Now, we have two ways to find inverses: by the Extended Euclidean Algorithm and by the Fast Exponentiation Algorithm.

2. Compute $2^{56} \% 1001$ using the Fast Exponentiation Algorithm.

X	e	Y
2	56	1
4	28	1
16	14	1
256	7	1
256	6	256
471	3	256
471	2	456

620	1	456
620	0	438

So, $2^{56}\%1001 = 438$